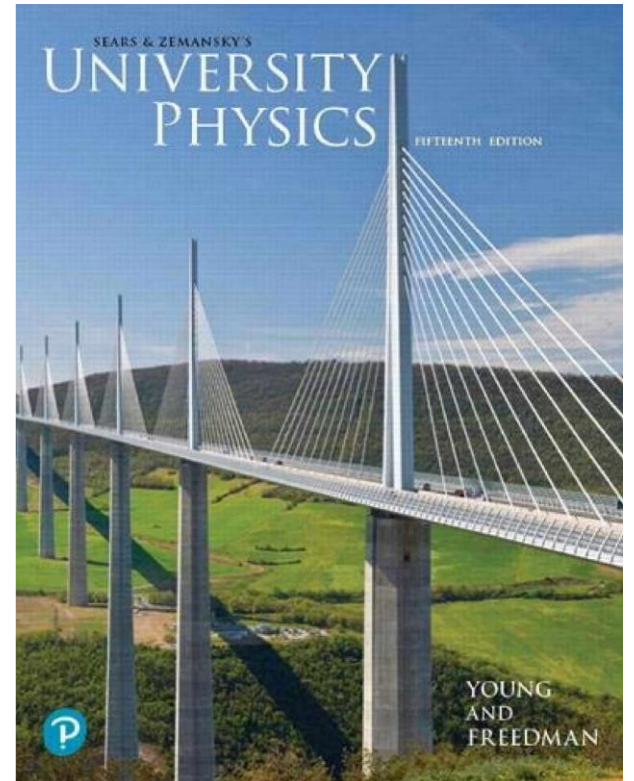


普通物理I PHYS1181

流体力学

Fluid mechanics



本节课（流体力学）：

A. 流体基本概念

B. 流体静力学

帕斯卡定律

阿基米德定律与浮力

C. 流体动力学

连续性方程

伯努利方程

托里拆利定理

D. *粘滞流体

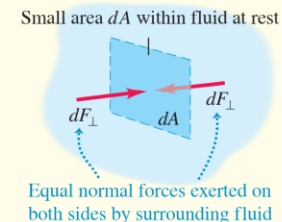
CHAPTER 12 SUMMARY

Density and pressure: Density is mass per unit volume. If a mass m of homogeneous material has volume V , its density ρ is the ratio m/V . Specific gravity is the ratio of the density of a material to the density of water. (See Example 12.1.)

Pressure is normal force per unit area. Pascal's law states that pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid. Absolute pressure is the total pressure in a fluid; gauge pressure is the difference between absolute pressure and atmospheric pressure. The SI unit of pressure is the pascal (Pa): $1 \text{ Pa} = 1 \text{ N/m}^2$. (See Example 12.2.)

$$\rho = \frac{m}{V} \quad (12.1)$$

$$p = \frac{dF_{\perp}}{dA} \quad (12.2)$$



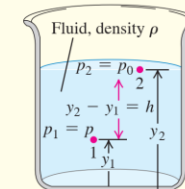
Pressures in a fluid at rest: The pressure difference between points 1 and 2 in a static fluid of uniform density ρ (an incompressible fluid) is proportional to the difference between the elevations y_1 and y_2 . If the pressure at the surface of an incompressible liquid at rest is p_0 , then the pressure at a depth h is greater by an amount ρgh . (See Examples 12.3 and 12.4.)

$$p_2 - p_1 = -\rho g(y_2 - y_1) \quad (12.5)$$

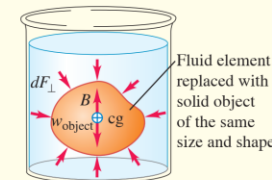
(pressure in a fluid of uniform density)

$$p = p_0 + \rho gh \quad (12.6)$$

(pressure in a fluid of uniform density)



Buoyancy: Archimedes's principle states that when an object is immersed in a fluid, the fluid exerts an upward buoyant force on the object equal to the weight of the fluid that the object displaces. (See Example 12.5.)



Fluid flow: An ideal fluid is incompressible and has no viscosity (no internal friction). A flow line is the path of a fluid particle; a streamline is a curve tangent at each point to the velocity vector at that point. A flow tube is a tube bounded at its sides by flow lines. In laminar flow, layers of fluid slide smoothly past each other. In turbulent flow, there is great disorder and a constantly changing flow pattern.

Conservation of mass in an incompressible fluid is expressed by the continuity equation, which relates the flow speeds v_1 and v_2 for two cross sections A_1 and A_2 in a flow tube. The product Av equals the volume flow rate, dV/dt , the rate at which volume crosses a section of the tube. (See Example 12.6.)

Bernoulli's equation states that a quantity involving the pressure p , flow speed v , and elevation y has the same value anywhere in a flow tube, assuming steady flow in an ideal fluid. This equation can be used to relate the properties of the flow at any two points. (See Examples 12.7–12.10.)

$$A_1 v_1 = A_2 v_2 \quad (12.10)$$

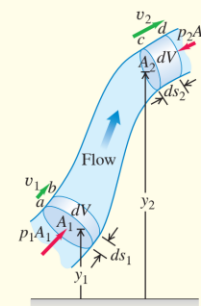
(continuity equation, incompressible fluid)

$$\frac{dV}{dt} = Av \quad (12.11)$$

(volume flow rate)

$$p + \rho gy + \frac{1}{2} \rho v^2 = \text{constant} \quad (12.18)$$

(Bernoulli's equation)



A. 流体 (fluid)

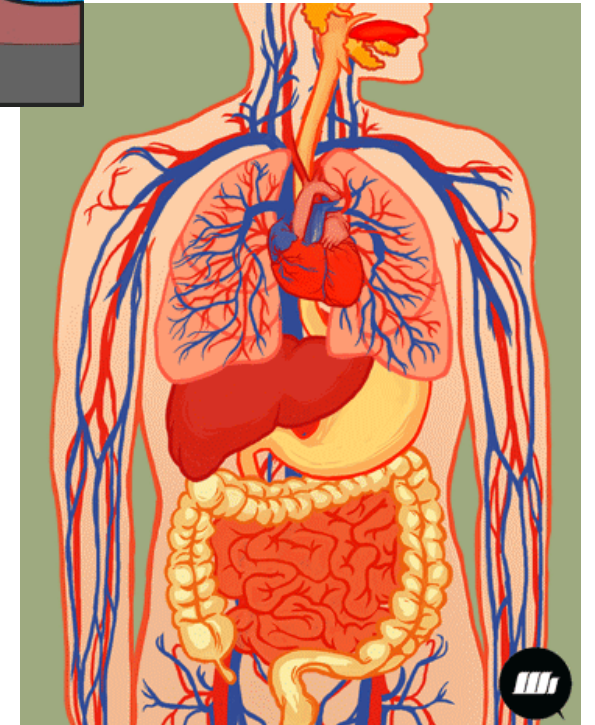
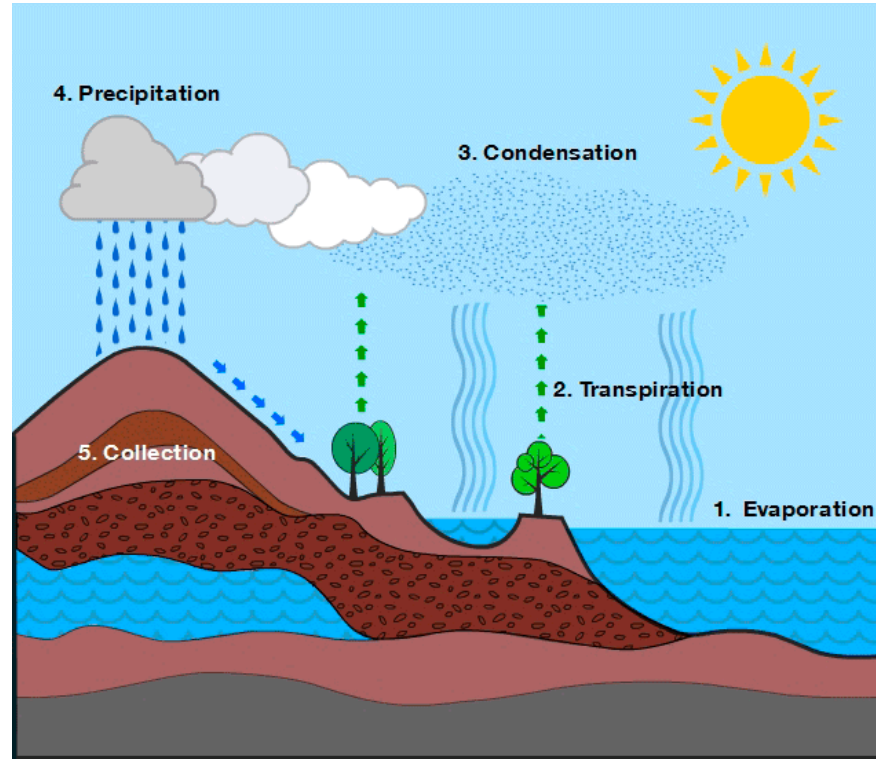
A1: 什么是流体?

A **fluid** is any substance that can flow and change the shape of the volume that it occupies.

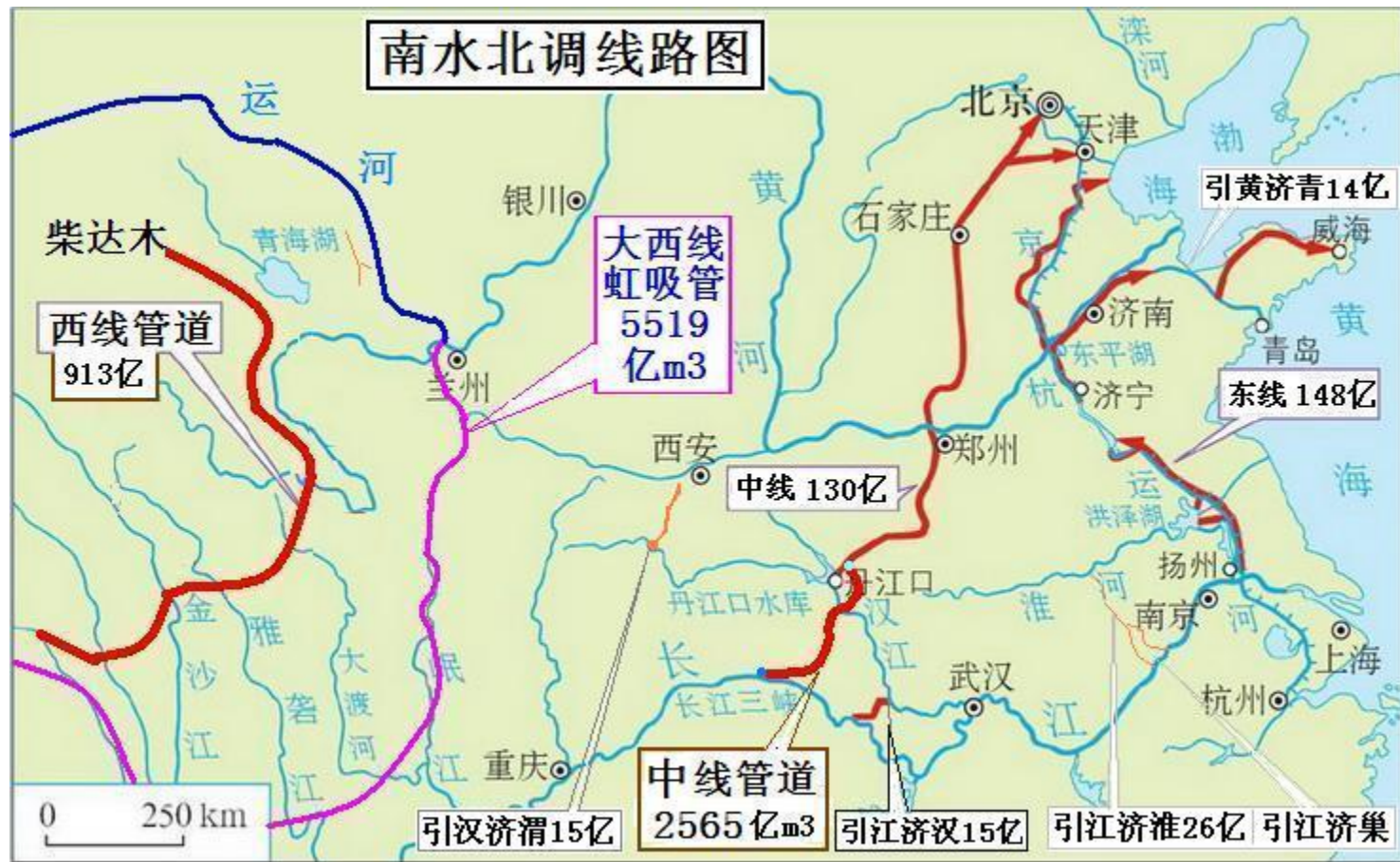
(By contrast, a solid tends to maintain its shape.)

航空航天、水利气象、海洋科学
交通运输、能源发电、生命科学
大禹治水、南水北调、三峡工程、。。。
——流体力学无处不在，已成为除固体力学外研究最多、
也是最为重要的力学分支

流体力学包括：流体静力学、流体动力学



中国古代繁荣富强的秘密：善于治水



南水北调工程优化了水资源配置。工程累计调水量近300亿立方米，直接受益人口超过1.2亿人，成为多个重要城市生活用水的主力水源。南水北调为**京津冀协同发展、雄安新区建设**等国家重大战略实施提供了可靠的水资源支撑。“南水”占到**北京中心城区自来水供水量七成以上**，天津市14个行政区全部用上“南水”，河南多个城市主城区100%使用“南水”，河北石家庄、邯郸、沧州等市的90多个县区受益，提升了江苏苏中苏北地区防洪排涝抗旱能力。

A2: 气体、液体、密度

流体主要包括：气体 (gases) 和液体 (liquids)

密度定义：

$$\rho = \frac{\Delta m}{\Delta V}.$$

对于均匀物质：

Density of a homogeneous material $\rho = \frac{m}{V}$
Mass of material
Volume occupied by material



常见物质的密度

Material or Object	Density (kg/m ³)
Interstellar space	10 ⁻²⁰
Best laboratory vacuum	10 ⁻¹⁷
Air: 20°C and 1 atm pressure	1.21
20°C and 50 atm	60.5
Styrofoam	1 × 10 ²
Ice	0.917 × 10 ³
Water: 20°C and 1 atm	0.998 × 10 ³
20°C and 50 atm	1.000 × 10 ³
Seawater: 20°C and 1 atm	1.024 × 10 ³
Whole blood	1.060 × 10 ³
Iron	7.9 × 10 ³
Mercury (the metal, not the planet)	13.6 × 10 ³
Earth: average	5.5 × 10 ³
core	9.5 × 10 ³
crust	2.8 × 10 ³
Sun: average	1.4 × 10 ³
core	1.6 × 10 ⁵
White dwarf star (core)	10 ¹⁰
Uranium nucleus	3 × 10 ¹⁷
Neutron star (core)	10 ¹⁸

B. 流体静力学

B1: 流体静力学: 压强

流体静止时，不存在任何切向力（否则就会流动），
所以我们可以得到：

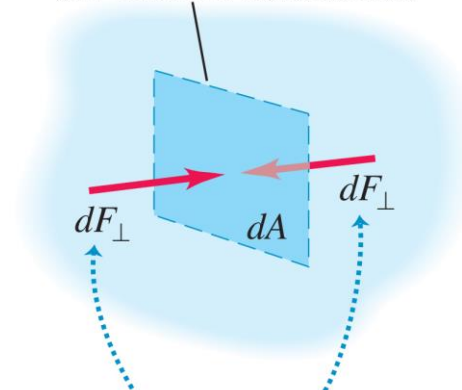
流体静力学第一结论：流体内部压力总是垂直于流体内部任意一个面

流体中某一点处的压强等于：

$$\text{Pressure at a point in a fluid} \rightarrow p = \frac{dF_{\perp}}{dA}$$

Normal force exerted by fluid on a small surface at that point
Area of surface

A small surface of area dA within a fluid at rest



The surface does not accelerate, so the surrounding fluid exerts equal normal forces on both sides of it. (The fluid cannot exert any force parallel to the surface, since that would cause the surface to accelerate.)

若在该点所在的某一平面上处处压力均匀（比如重力作用下的流体中等高的平面），
则有：

$$p = \frac{F_{\perp}}{A}$$

压强是个标量，其单位：

$$1 \text{ pascal} = 1 \text{ Pa} = 1 \text{ N/m}^2$$

大气压：

$$(p_a)_{\text{av}} = 1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$$
$$= 1.013 \text{ bar} = 1013 \text{ millibar}$$
$$= 760 \text{ torr} = 14.7 \text{ lb/in.}^2$$

流体静力学第二结论：流体内部某点处压强在任意方向上都相同

流体静力学第三结论：重力作用下的流体平衡方程

Pressure at depth h
in a fluid of uniform
density

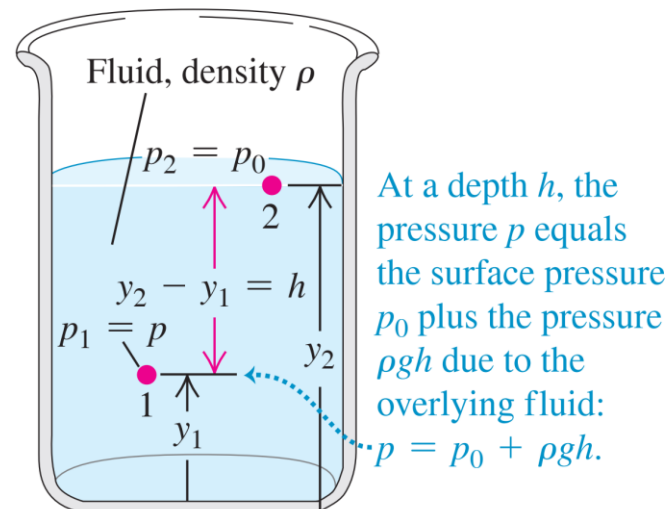
$$p = p_0 + \rho gh$$

Pressure at surface of fluid

Uniform density of fluid

Depth below surface

Acceleration due to gravity ($g > 0$)



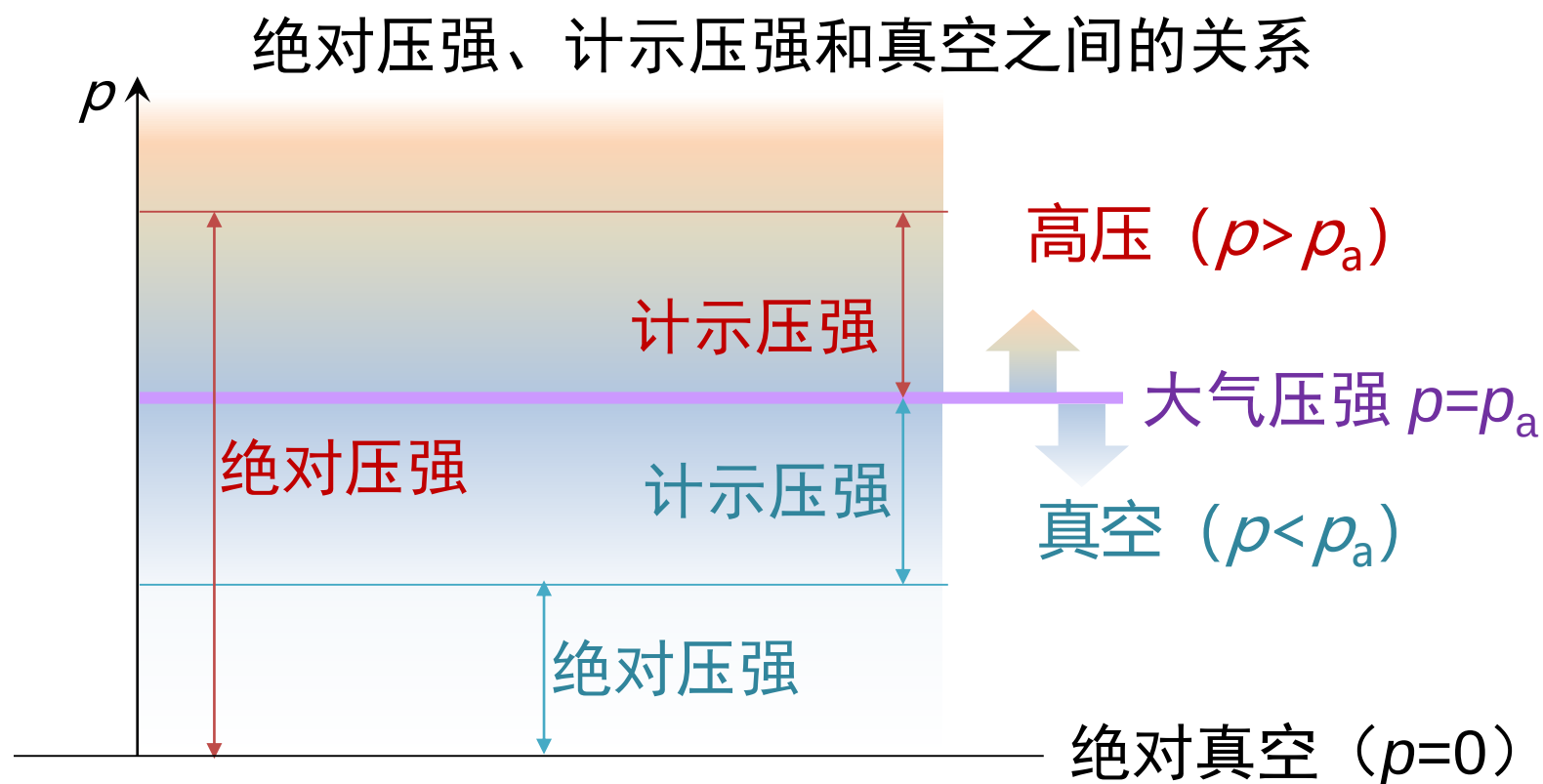
Pressure difference between levels 1 and 2:

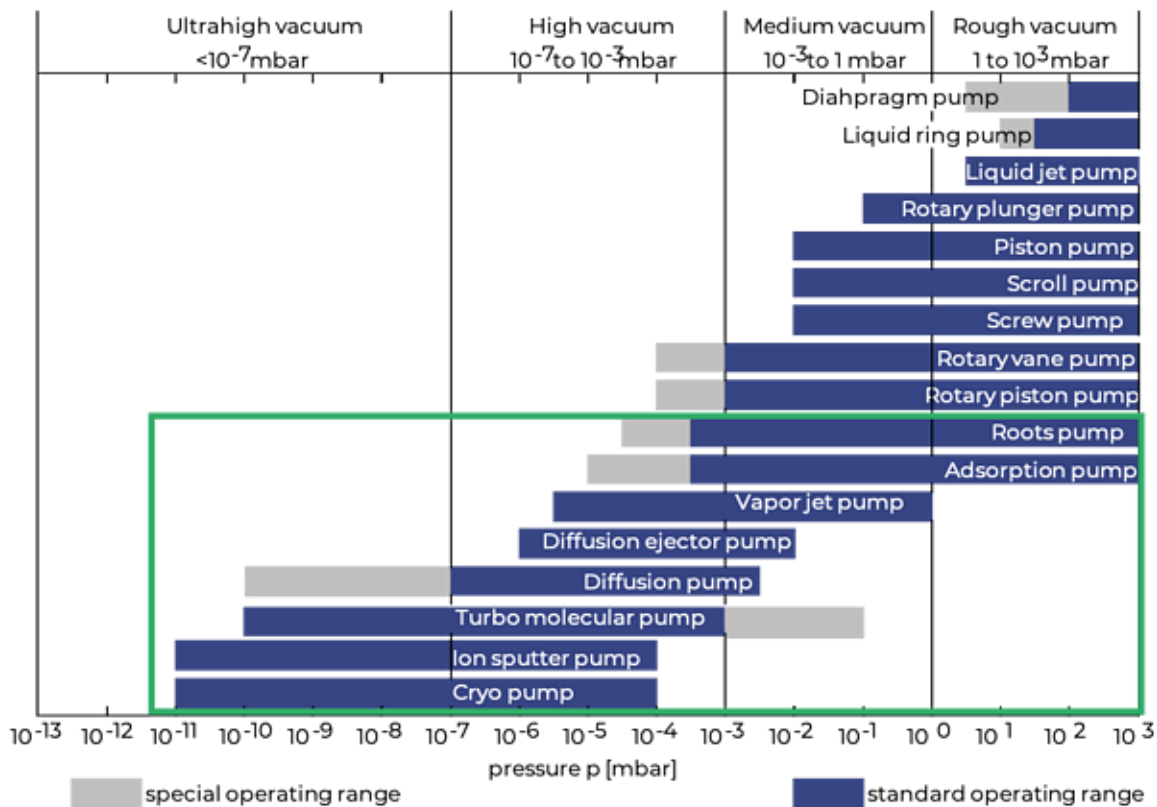
$$p_2 - p_1 = -\rho g(y_2 - y_1)$$

The pressure is greater at the lower level.

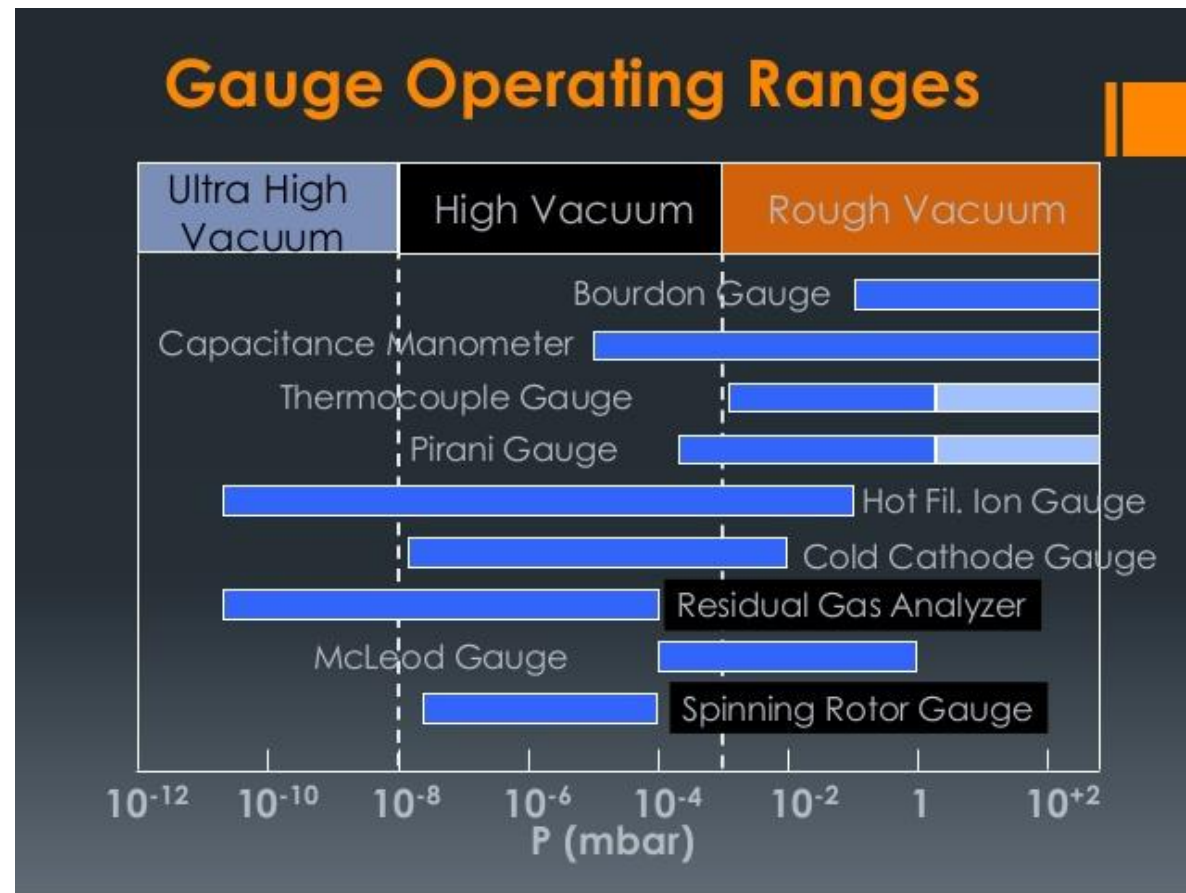
B2: 压强的测量

- 绝对压强 (absolute pressure)
- 计示压强 (gauge pressure) (表压)





真空的获取：真空泵



真空度（压强）的测量：真空规

B3: 帕斯卡原理 (Pascal's principle)



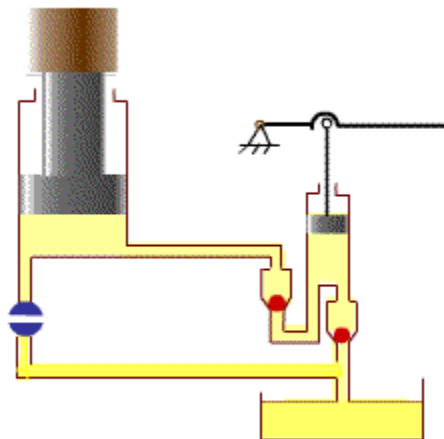
A change in the pressure applied to an enclosed incompressible fluid is transmitted undiminished to every portion of the fluid and to the walls of its container.

帕斯卡原理： 封闭容器中的不可压缩流体的某一部分施加的压强变化，将毫无损失地传递至流体的各个部分和容器壁。

根据帕斯卡原理，在液体系统中的一个活塞上施加一定的压强，必将在另一个活塞上产生相同的压强增量。如果第二个活塞的面积是第一个活塞的面积10倍，那么作用于第二个活塞上的力将增大为第一个活塞的10倍，而两个活塞上的压强仍然相等。



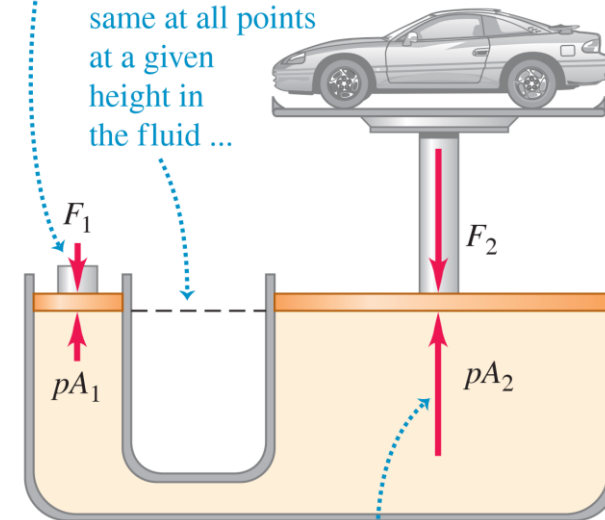
Heimlich maneuver
(海姆利克操作法)



hydraulic lift (液压千斤顶)

A small force is applied to a small piston.

Because the pressure p is the same at all points at a given height in the fluid ...



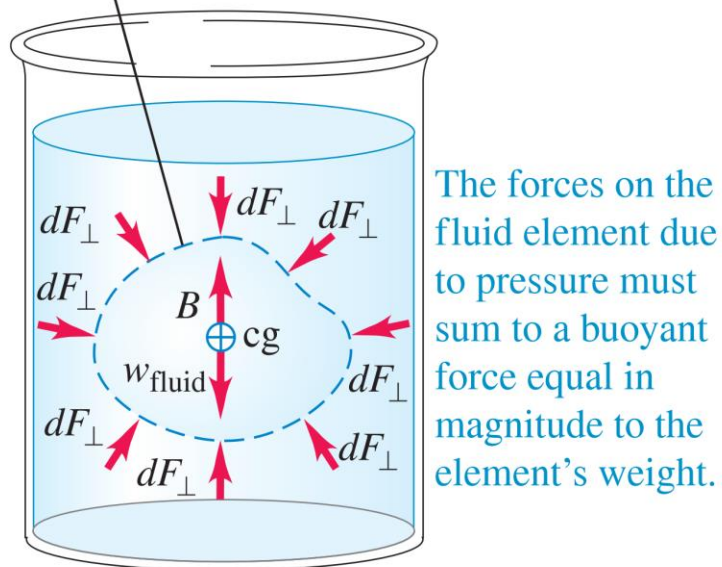
... a piston of larger area at the same height experiences a larger force.

B4: 阿基米德原理 (Archimedes' principle) 与浮力

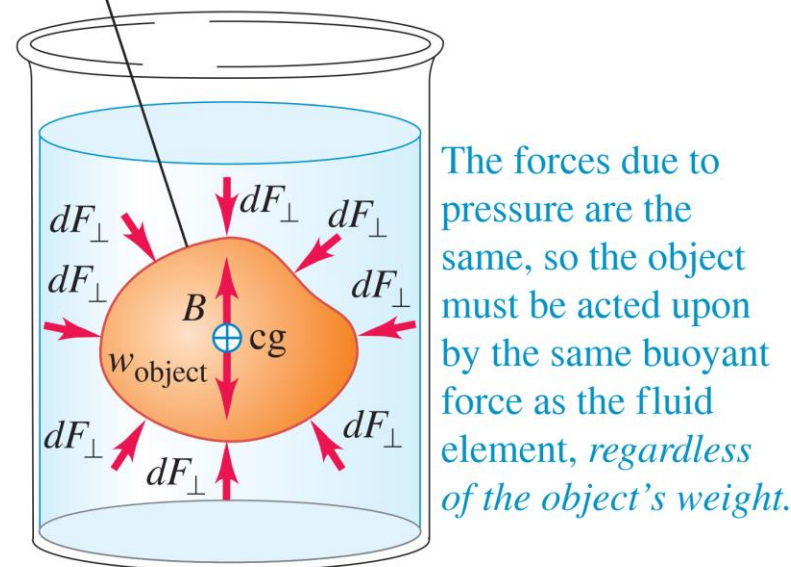
When a body is fully or partially submerged in a fluid, a buoyant force $\vec{F}_b$ from the surrounding fluid acts on the body. The force is directed upward and has a magnitude equal to the weight $m_f g$ of the fluid that has been displaced by the body.

阿基米德原理： 浸在流体中的物体受到竖直向上的浮力，其大小等于物体所排开流体的重力。

(a) Arbitrary element of fluid in equilibrium



(b) Fluid element replaced with solid object of the same size and shape



C. 流体力学

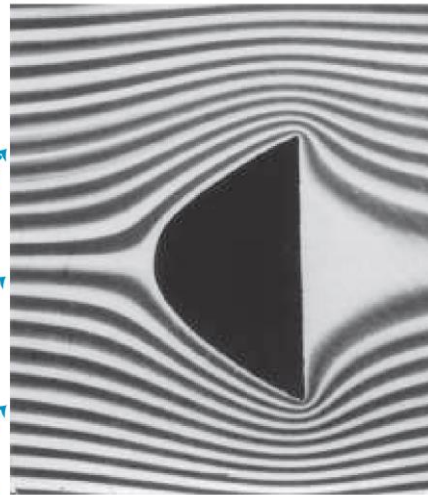
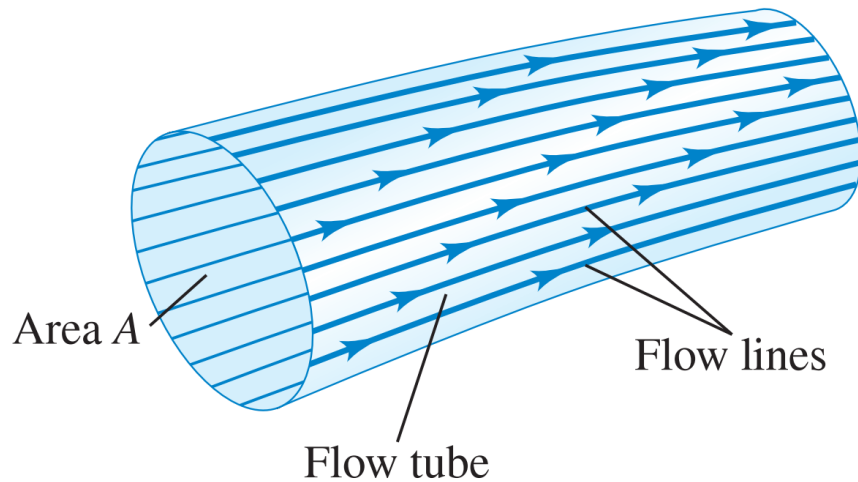
真实流体的运动（流动）极为复杂，并且尚未完全得到理解。因此，我们在研究流体运动的时候，先讨论最简单的**理想流体**的流动，主要有四个假设：

- 1、定常流动（任一点流速不随时间发生变化）
- 2、不可压缩
- 3、无粘性
- 4、无旋转

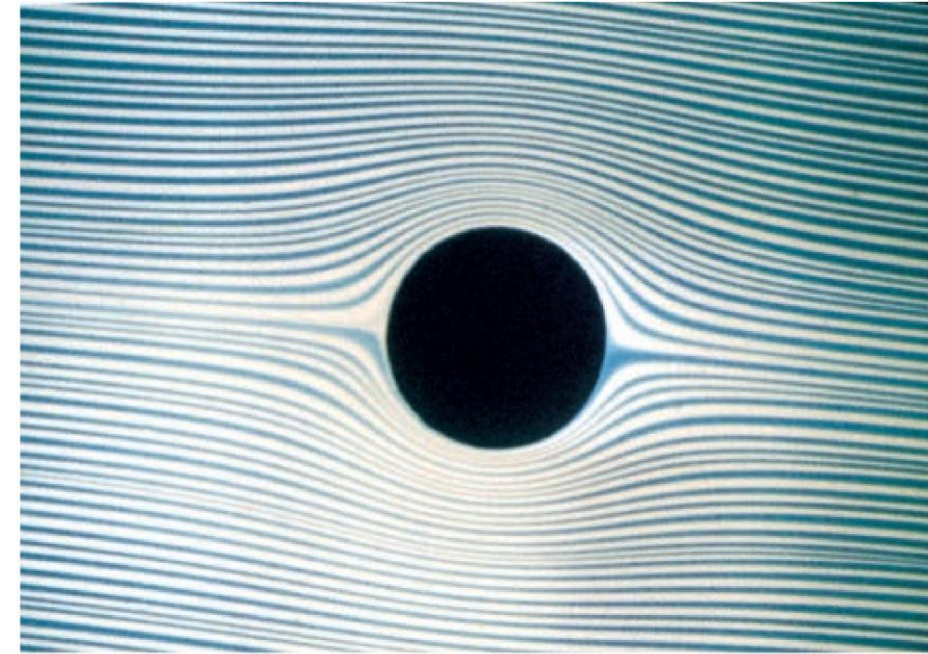


C2: 流管、流线 (flow tube, flow lines)

Figure 12.18 A flow tube bounded by flow lines. In steady flow, fluid cannot cross the walls of a flow tube.



Dark-colored dye follows streamlines of laminar flow (flow is from left to right).



Courtesy D. H. Peregrine, University of Bristol

汽车风洞实验



C2: 连续性方程 (continuity equation)

前提：不可压缩 → 密度不变

+

物质守恒 → 连续性方程

$$\rho A_1 v_1 dt = \rho A_2 v_2 dt$$

连续性方程：

Continuity equation
for an **incompressible fluid**:

$$A_1 v_1 = A_2 v_2$$

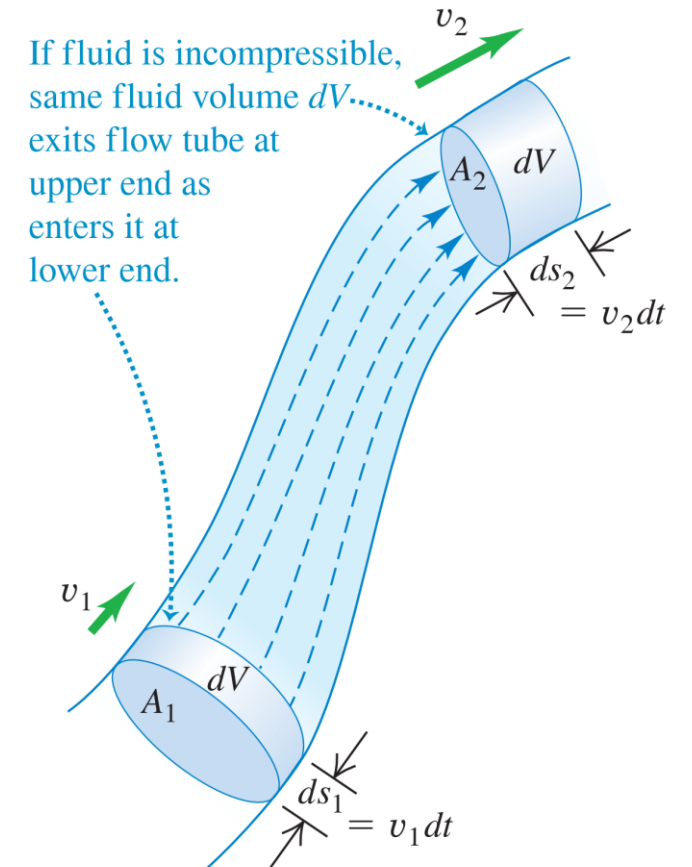
Cross-sectional area of flow tube at two points (see Fig. 12.21)
Speed of flow at the two points

体积流速：

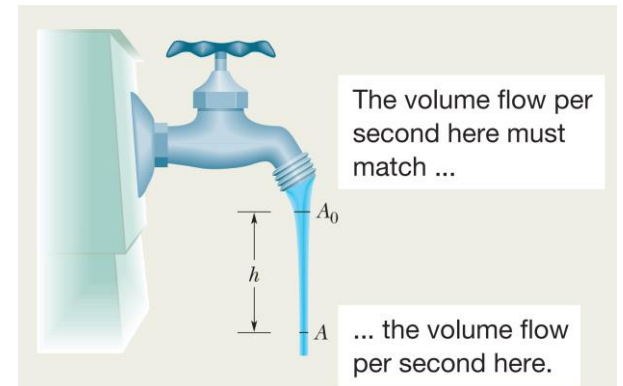
$$R_V = Av = \text{a constant} \quad (\text{volume flow rate, equation of continuity}),$$

质量流速：

$$R_m = \rho R_V = \rho Av = \text{a constant} \quad (\text{mass flow rate}).$$



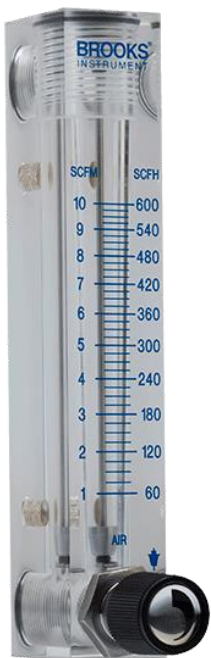
If fluid is incompressible, product Av (tube area times speed) has same value at all points along tube.



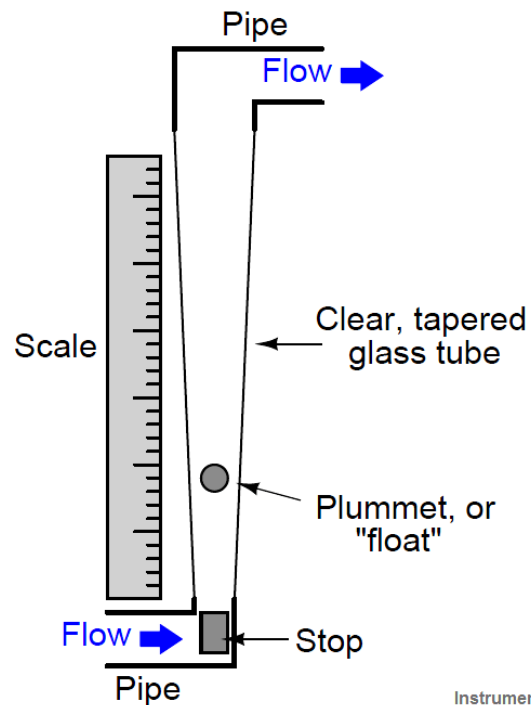
实际应用中，由于流体压强是可变的，流速往往不是常数，需要测量和控制流量（单位时间内的流量）：

浮子流量计（可变面积流量计、**Rotameter**）可以用来测量流速，是最简单快捷的流量测量计。

在工业加工和科研中，稳定流速的气源非常重要，例如芯片刻蚀、化学化工、晶体生长等。因此，需要高精度的质量流量计（**Mass flow controller, MFC**）



Rotameter



InstrumentationTools.com

chem17.com



C3: 伯努利方程 (Bernoulli equation)

根据能量守恒，压强对流体做的功： $dW = dK + dU$

其中：

$$\begin{cases} dW = p_1 A_1 ds_1 - p_2 A_2 ds_2 = (p_1 - p_2) dV \\ dK = \frac{1}{2} \rho dV (v_2^2 - v_1^2) \\ dU = \rho dV g (y_2 - y_1) \end{cases}$$

$dV = A_1 ds_1 = A_2 ds_2$

➡ $p_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = p_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$

可得：伯努利方程：

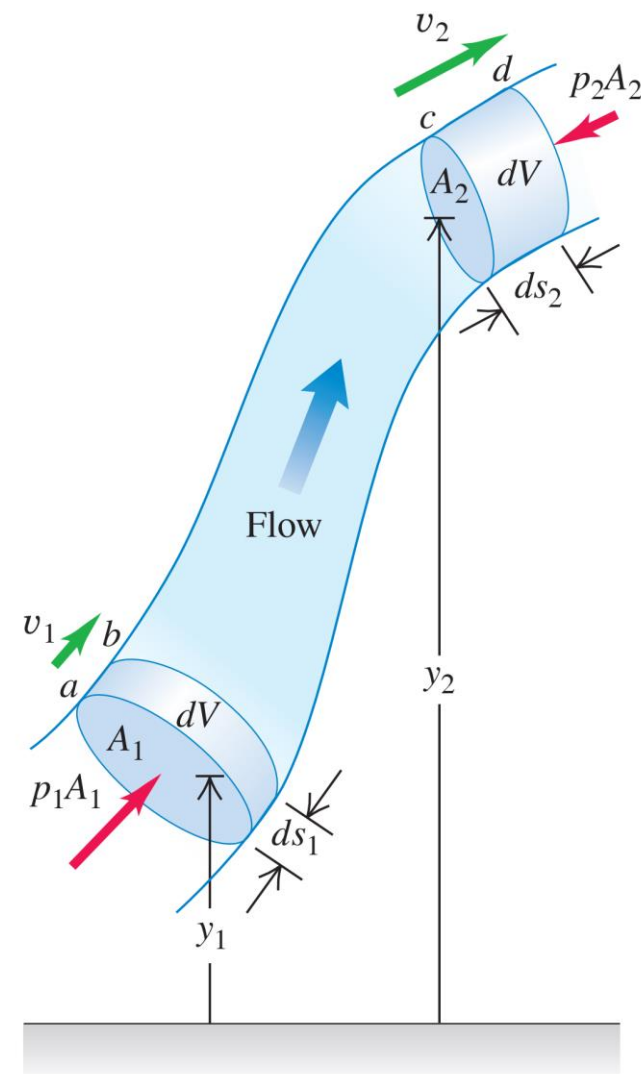
Bernoulli's equation
for an ideal,
incompressible fluid:

Pressure Fluid density Value is **same** at all points in flow tube.

$$p + \rho g y + \frac{1}{2} \rho v^2 = \text{constant}$$

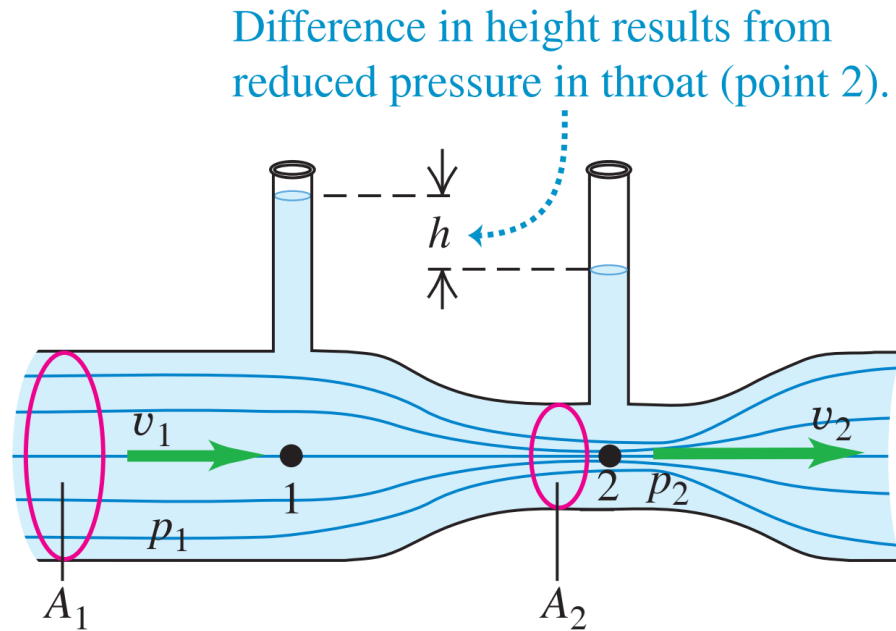
Acceleration due to gravity Elevation Flow speed

Figure 12.23 Deriving Bernoulli's equation. The net work done on a fluid element by the pressure of the surrounding fluid equals the change in the kinetic energy plus the change in the gravitational potential energy.



C4: 伯努利方程的应用:

(a) Venturi meter (文丘里流量计) :



$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

$$v_2 = (A_1/A_2)v_1$$

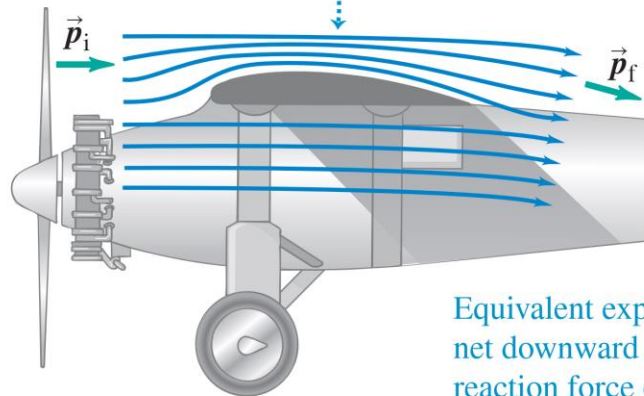
$$p_1 - p_2 = \frac{1}{2}\rho v_1^2 \left[\left(\frac{A_1}{A_2} \right)^2 - 1 \right]$$

$$v_1 = \sqrt{\frac{2gh}{(A_1/A_2)^2 - 1}}$$

(b) 飞机机翼的升力:

(a) Flow lines around an airplane wing

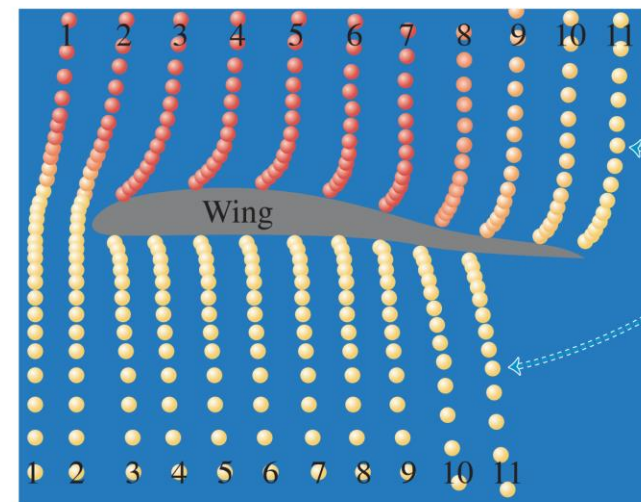
Flow lines are crowded together above the wing, so flow speed is higher there and pressure is lower.



Equivalent explanation: Wing imparts a net downward momentum to the air, so reaction force on airplane is upward.



(b) Computer simulation of air parcels flowing around a wing, showing that air moves much faster over the top than over the bottom

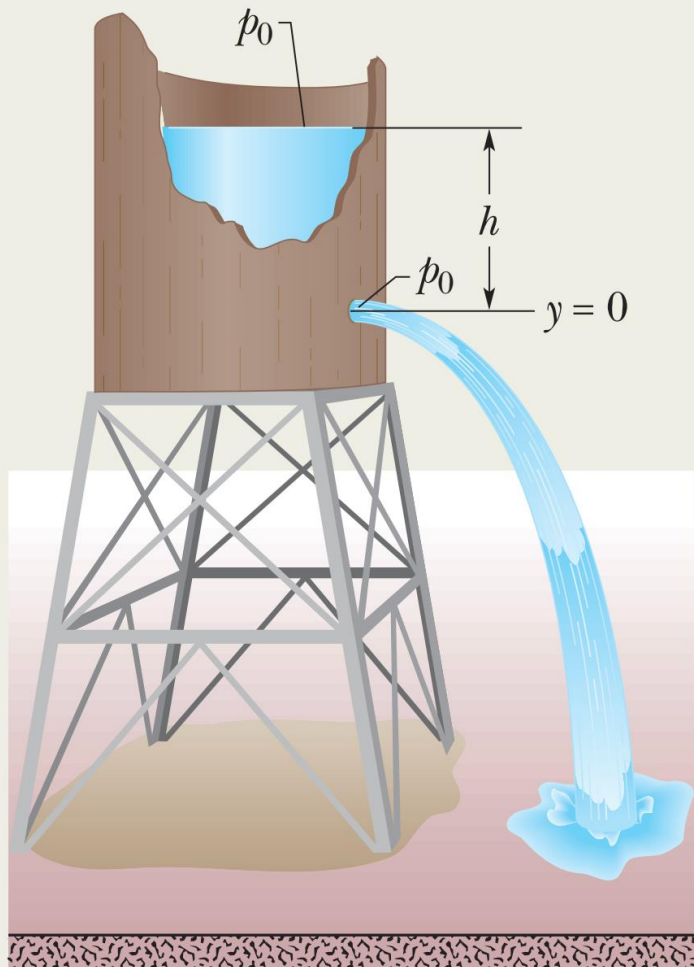


Notice that air particles that are together at the leading edge of the wing do *not* meet up at the trailing edge!

*实际上伯努利原理只是飞机产生升力的部分原因。机翼的具体形状、攻角等都对飞机的升力有作用。

(c) 托里拆利定理:

小孔面积 a , 出水速度 v , 水箱截面积 A , 水位下降速度 v_0



$$R_V = av = Av_0$$

$$v_0 = \frac{a}{A} v.$$

$$a \ll A, v_0 \ll v.$$

$$p_0 + \frac{1}{2}\rho v_0^2 + \rho gh = p_0 + \frac{1}{2}\rho v^2 + \rho g(0).$$

因此, 小孔出水速度: $v = \sqrt{2gh}.$

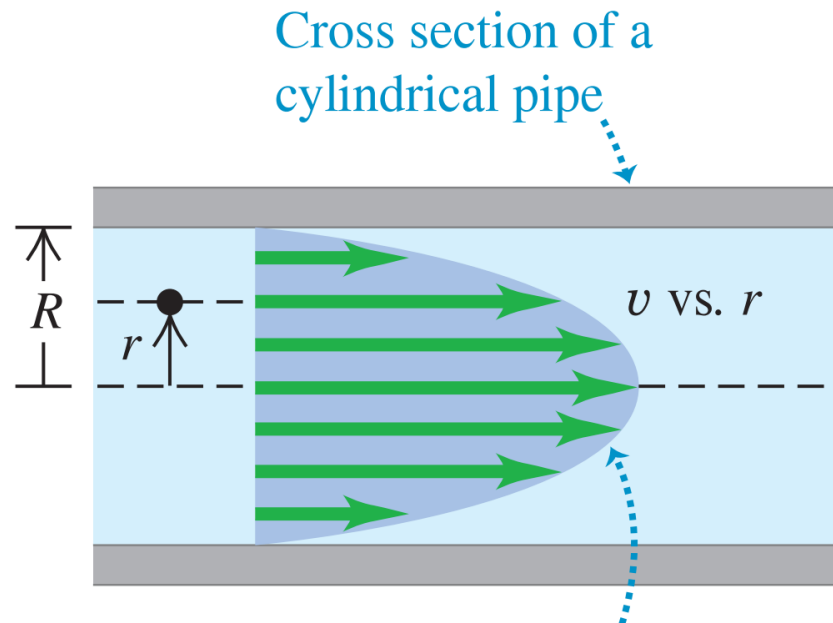
*D. 粘滯流体

实际的流体都具有粘滯性，即使是水、空气

典型的粘滯流体的例子：

鸡蛋清、麦芽糖、蜂蜜

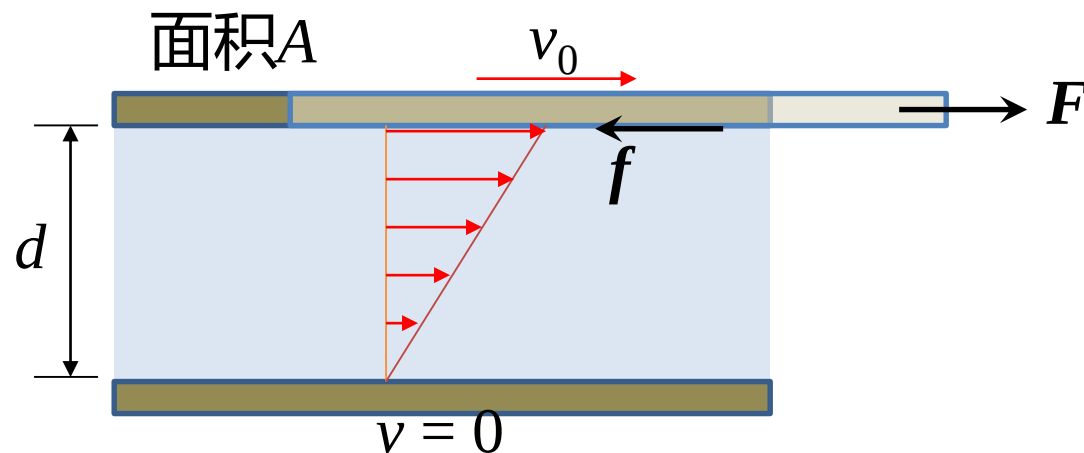
岩浆、柏油、胶水



The velocity profile for viscous fluid flowing in the pipe has a parabolic shape.



D1: 粘滞系数



考虑实验：两块间距为 d 中间夹有流体的固体板面，保持下板固定，以拉力 F 水平拉上板使其以低速 v_0 直线运动。

上板除了受到拉力 F 外，还受到液体的粘滞阻力 f ，大小相等，方向相反。

对于**牛顿流体**，粘滞力与速度成正比，有牛顿粘滞定律：

粘滞力 f 的大小与板的面积 A 、 v_0/d ：

$$\frac{f}{A} = \eta \frac{v_0}{d}$$

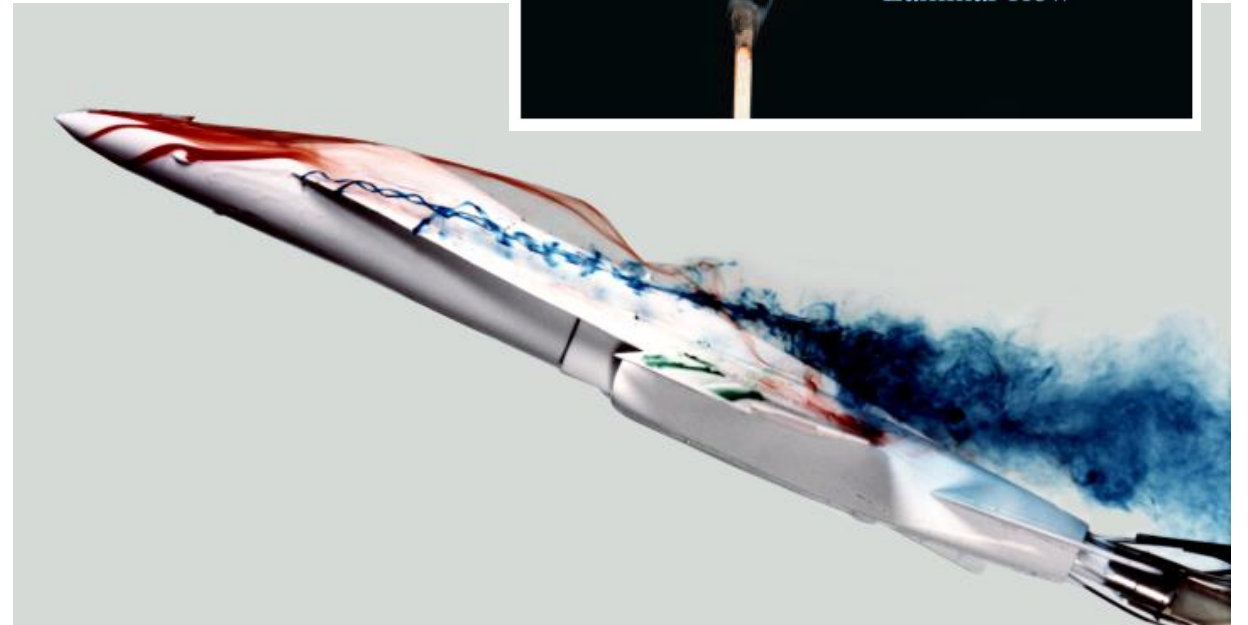
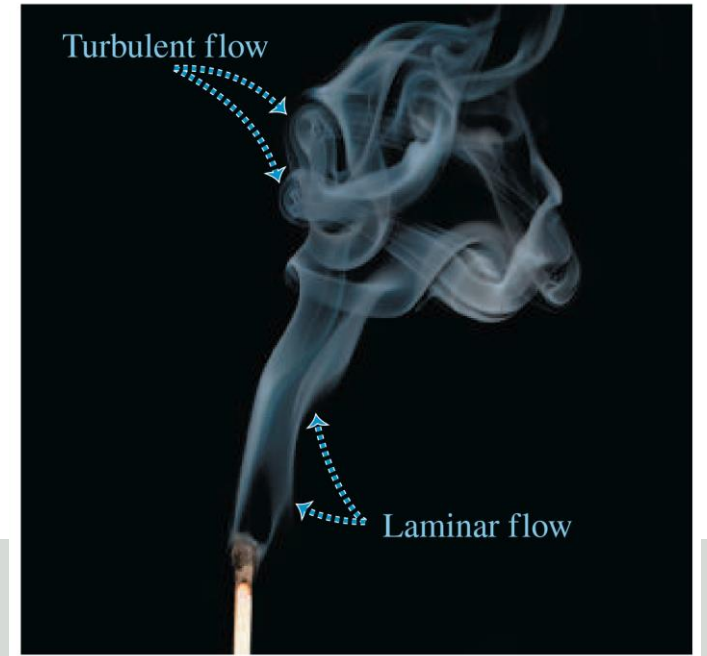
不符合牛顿粘滞定律的流体为非牛顿流体。

D2: 层流与湍流 (laminar flow, turbulent flow)

(a) Low speed:
laminar flow



(b) High speed:
turbulent flow



萌萌战队

萌萌战队 bilibili

一位茶壶玩家说他看直播时